Complete Empirical Capture of the Agent/Environment Interaction across Phylogeny, Discipline, and Spatiotemporal Scale

FreeMoCap Foundation, Inc.

Topic: Scientific Instrumentation for Sensing and Imaging

1. Mission

- **Funnel Statement**: **[Reductionist's Promise]** The modern landscape of science is one of loosely connected pockets of hyper specialization within siloed disciplines and domains of inquiry.
- **Hedge Statement**: While this approach has led to many advances over the past century, the current model of specialized single-domain advancement has failed to create integrative frameworks to combine the individual threads of single-modality advancement into an coherent whole.
- **Gap Statement**: Perverse incentives and structural skill-ceilings of academia results in a system fundamentally incapable of building sophisticated tool capable of managing the complexity of true interdisciplinary research.
- **Hero Statement**: A high quality integrated platform directly designed for the use-case development of complete empirical capture of an awake behaving human-and-non-human-animals (HANHA) would transform the scientific landscape by creating a technical commons whereby specialists and cross-disciplnary researchers can productively interact in serivce of the shared and unified scientific endeavor.

MISSION STATEMENT - Build a Platform for **Complete Empirical Capture of the Agent/Environment Interaction** to fulfill the **Reductionist's Promise** and enable a **Mesoscale Revolution**

KEY DELIVERABLE - Scale-free Empirical Capture Volume Defined as **densely instrumented** 3d region wherein we can extract empirically grounded heterogenous sensor data into a coherent multi-domain estimate of all relevant internal and external variables that are relevant to the sensorimotor control over all timescales (sub-second reactions to longitudinal development).

ACTIONABLE TARGET - Align perceptuomotor neuroscience with legged robotics Eventual goal is "all of science especially biology", but for the timeline of this proposal we specifically target alignment between perceptuomotor neuroscience and legged robotics

1.1. Key Terms and Phrase

Complete Empirical Capture: Complete empirical capture means that we are recording every single possible thing that we can record about a given slice of time over continuous time.
Agent/Environment Interaction: Relates to the internal and external state of an actor (human, non-human animal, or artificial system).

1.2. Building a Scale-free Empirical Capture Volume

Defined as 3d region where in we can extract empirically grounded sensor data, spatiotemporally align, synchronize, and calibrate into a Coherent multi-domain estimate of all relevant internal and external variables that are relevant to the sensorimotor control over all timescales (sub-second reactions to longitudinal development).
Composable complexity - new sensors intergrate into base dataset

- Interspecies alignment though phylogenetic property Mapping
  - Any species with a Skull has the Skull property. If eye tracking data is not specified, eyes are fixed in head. If eye tracking is present, we pour emprically grounded estiamtes of measurable parameters (adduction/elevation). Eye models includes knowable but unmeasured values (like occular torsion or lens flexion), providing both future proofing of current research with optimistic preparation for future methodology which CAN measure those values

1.3. Enable Mesoscale research

- Cash in on reductionists dream by creating expansive framework for cross-domain integration of specialized insights
- Allow grounding of human/meso-scale research to reductionist micro-scale mechanisms
  - Still rewards specialized work
  - Provides context to guide specialized work in directions that help actionable meso-scale insights

1.3.1. [Specific Target] Align (perceptuomotor) neuroscience with (legged) robotics

- Recent robotics boom arising from high quality reinforcement learning frameworks (like Nvidia IsaacGym) alongside high quality inverse kinematics solvers (MuJoCo, OpenSim)
- Create our system in a format that can feed directly into those systems, so insights about and data from Humans and Animals can drive RL models for Robots, and inferred robotic control policies provide insight into biological behavior, perceptuomotor neural activity and musculoskeletal biomechanics.

1.4. Science as collaborative toolbuilding

- Through workshops and congresses, shift the culture of science towards an endeavor of shared collaborative tool building rather than an endless parade of 8-10 page pdfs

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2. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1–3]
  - etc
  - Ferret work
    -

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3. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

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3.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending 'mesoscale research revolution' to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1–3]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

5. Team Capabilities Statement

5.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

5.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

5.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.

- This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28,** 1224–1233 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18,** e1009575 (2022).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12,** e82410 (2023).